Operating Azure Applications

Understand telemetry, investigate failures, and respond to application health signals.

Azure Monitor • Application Insights • Guided lab

Publix Azure Foundational Workshop | Level 100–200

Opening: This module follows application hosting, messaging, and storage. Emphasize that production operations require visibility across the entire transaction, not just a single resource.

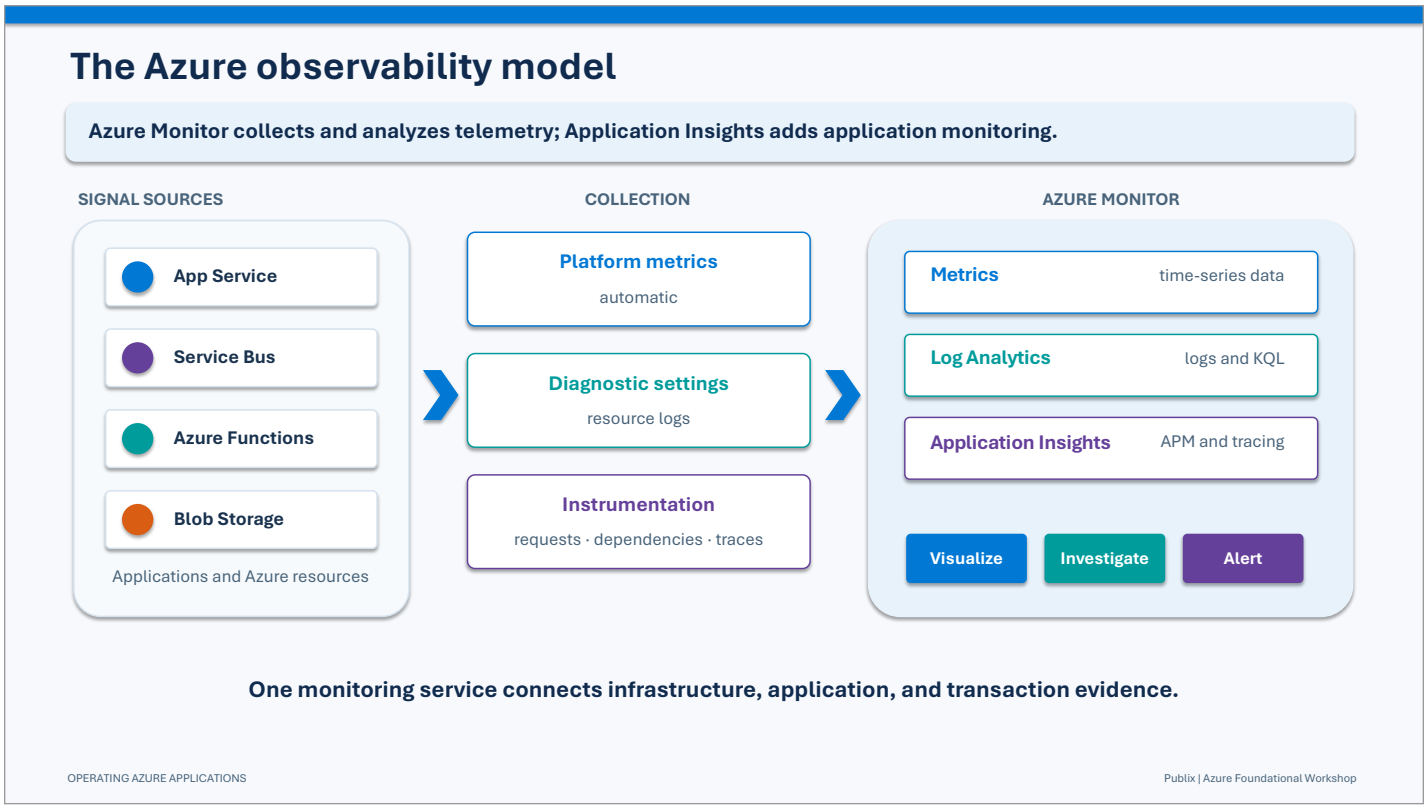
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview



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Learning objective is interpretation and basic troubleshooting, not deep KQL or production monitoring implementation.

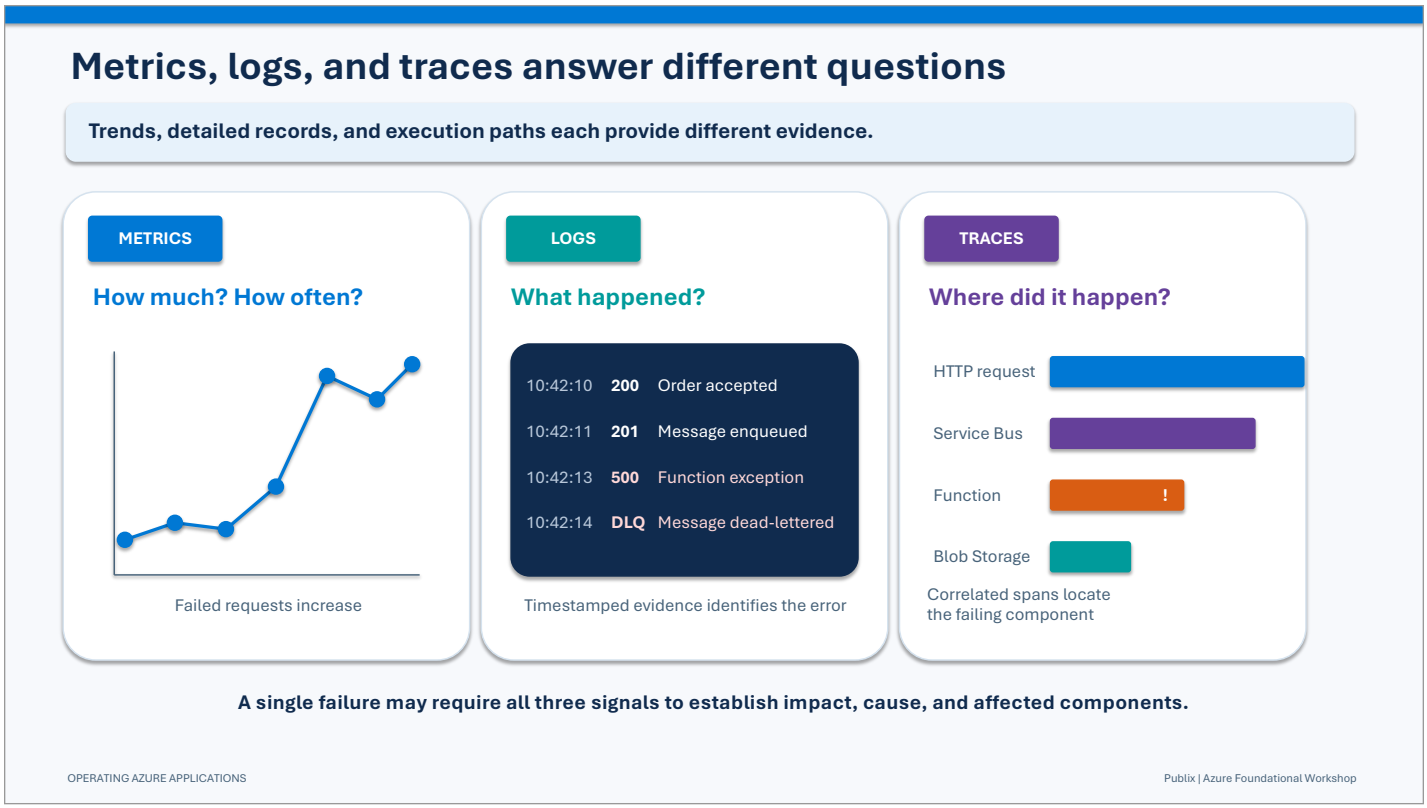
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- https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview



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Azure Monitor is Microsoft's unified observability service for collecting, analyzing, and acting on telemetry from cloud and hybrid environments. Application Insights is a feature of Azure Monitor focused on live application performance monitoring.

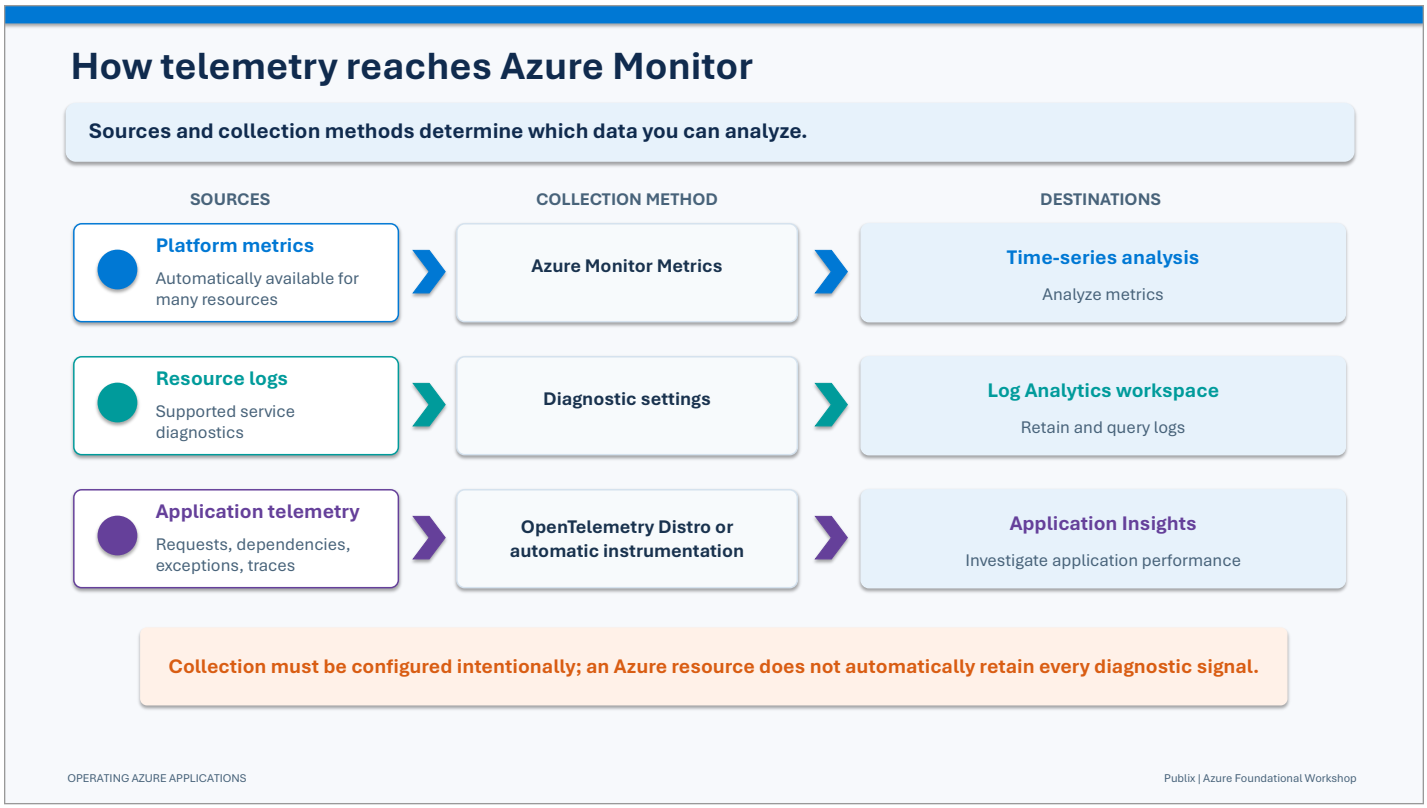
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- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview



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Explain Azure Monitor metrics versus log data. Distributed tracing is especially relevant to App Service -> Service Bus -> Functions workflows, but requires appropriate correlation and instrumentation.

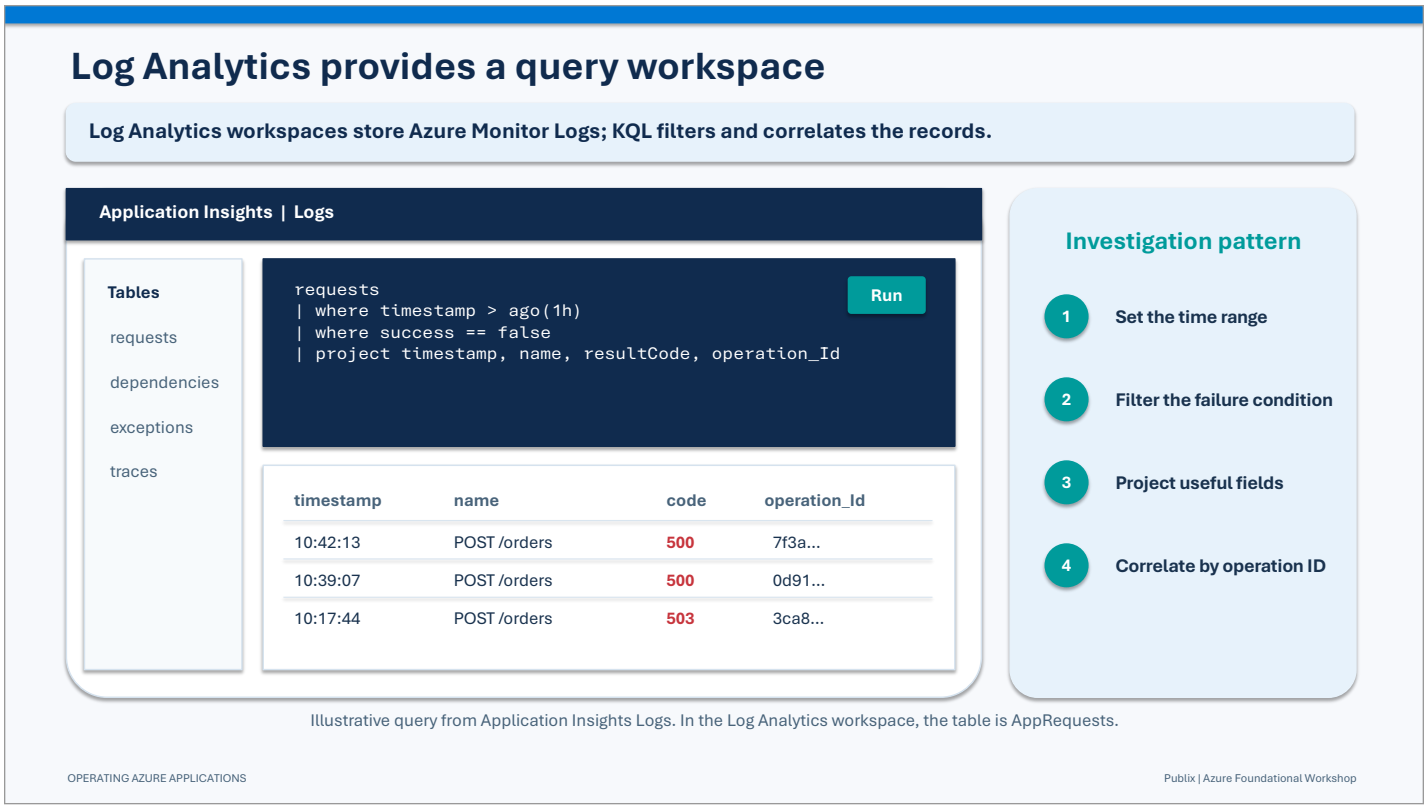
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/data-platform-metrics
- https://learn.microsoft.com/en-us/azure/azure-monitor/logs/data-platform-logs
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/data-model-complete



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Platform metrics are generally collected automatically. Resource logs often require diagnostic settings and destination selection. Application Insights needs instrumentation and configuration. Distinguish the signal's source from its destination. For application code, Microsoft recommends an OpenTelemetry Distro for most code-based apps (the Microsoft OpenTelemetry Distro for .NET, Node.js, and Python, or the Azure Monitor OpenTelemetry Distro for Java); some platforms also enable automatic instrumentation.

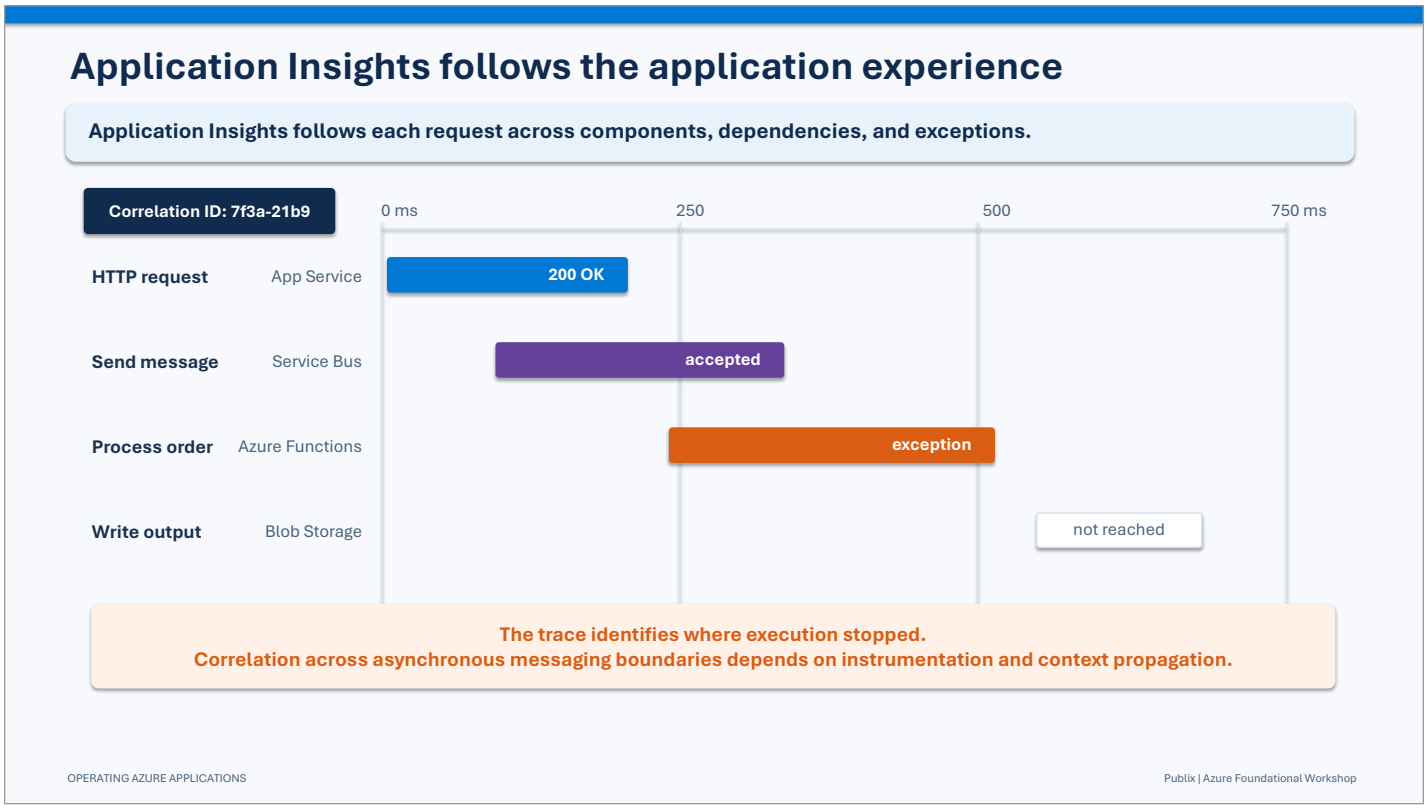
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/diagnostic-settings
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/opentelemetry-enable



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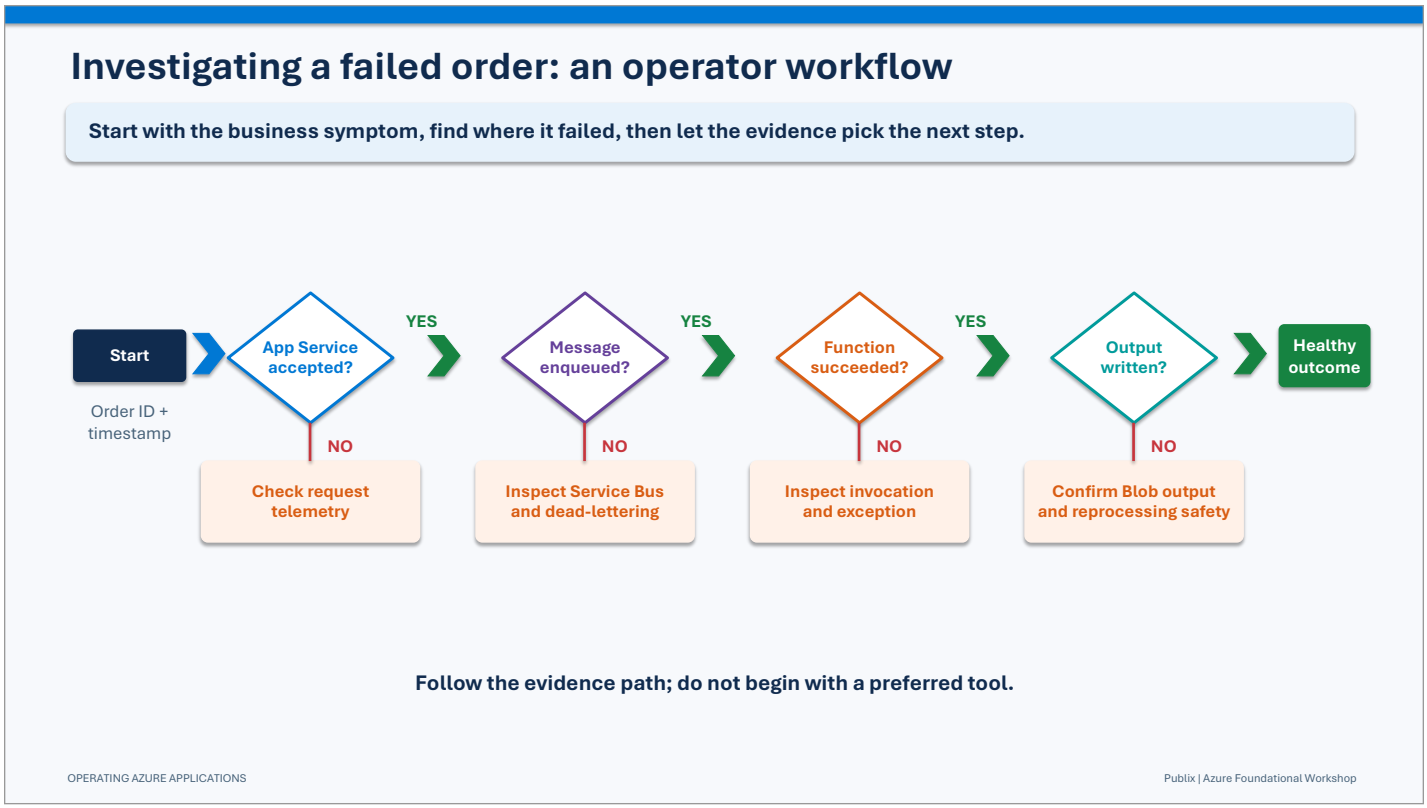
Run this query from the Application Insights Logs view, where the table is requests. In the Log Analytics workspace behind it, the same data is in AppRequests, with TimeGenerated instead of timestamp. Teach KQL concepts, not memorization.

[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/logs/log-analytics-overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/data-model-complete



Application Insights provides application performance monitoring. Explain request, dependency, exception, and distributed trace concepts. Service Bus processing correlation depends on instrumentation and context propagation; it is not automatic for every arbitrary workflow.

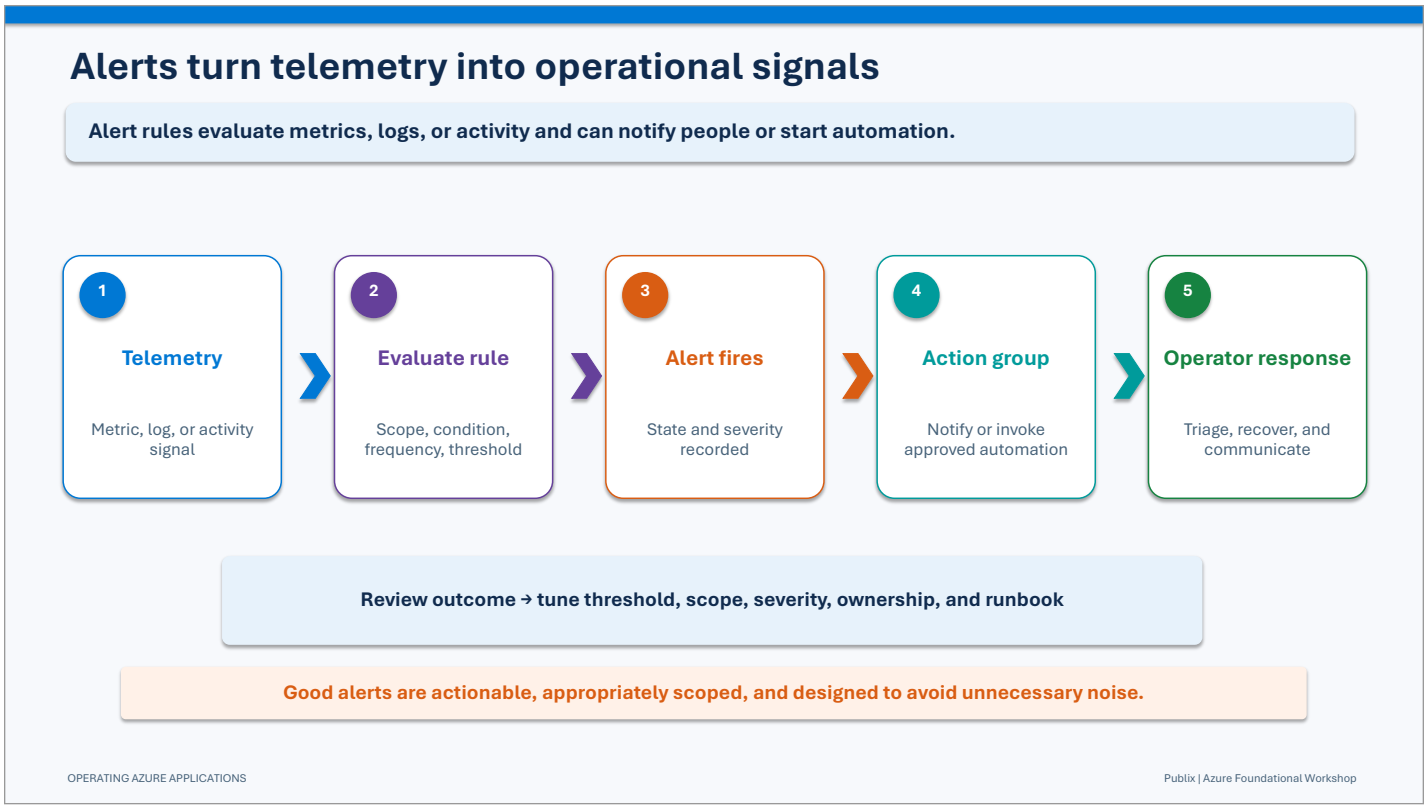
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/failures-performance-transactions



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Use a deliberately failed sample order from the Building Applications lab. Explain that correlation IDs must be included consistently in application and messaging telemetry. Distinguish a business validation failure from transient infrastructure failure.

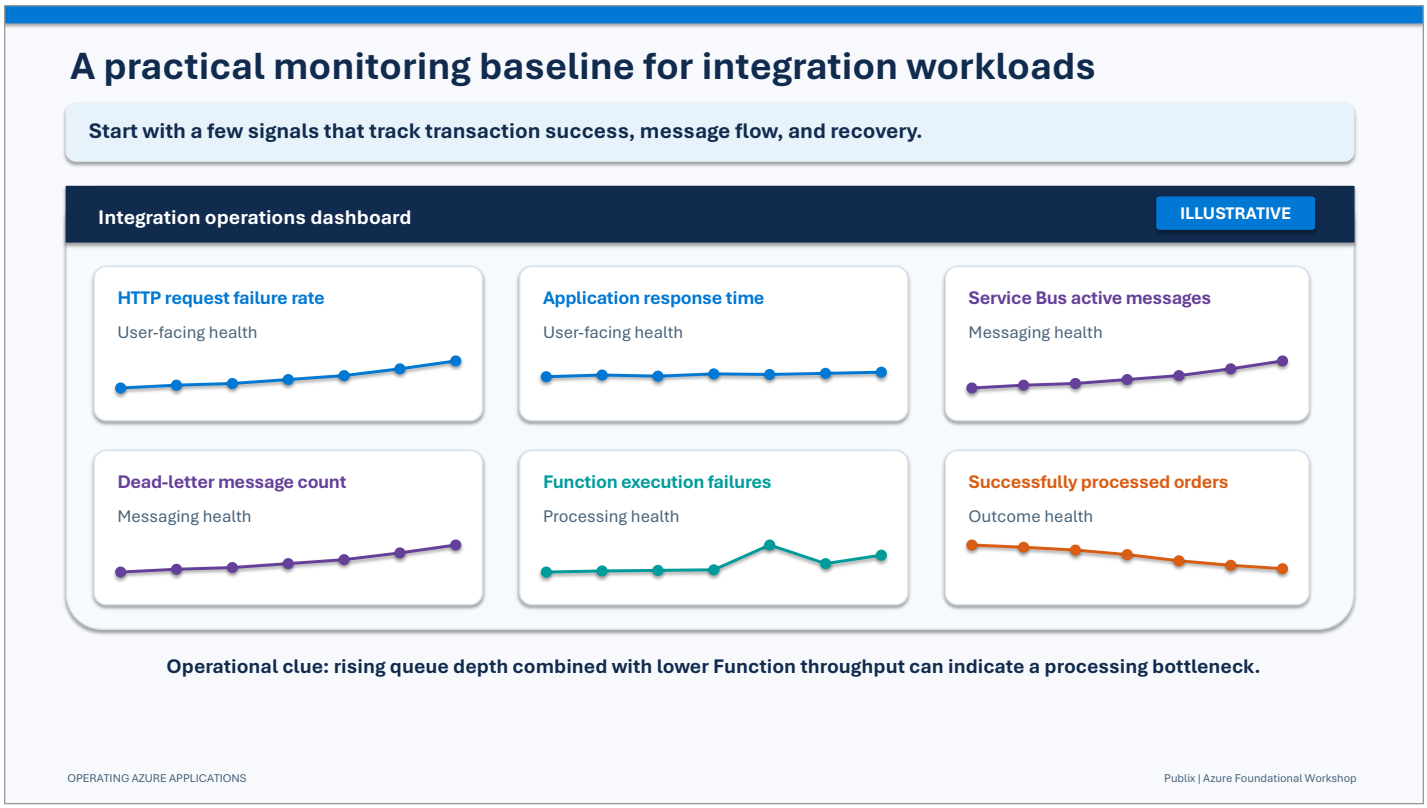
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/failures-performance-transactions
- https://learn.microsoft.com/en-us/azure/service-bus-messaging/service-bus-dead-letter-queues



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Azure Monitor alert rules can be based on metrics, logs, activity logs, and other signals. Action groups configure notifications and automation. Do not imply every Service Bus metric is directly alertable without confirming the selected metric and resource.

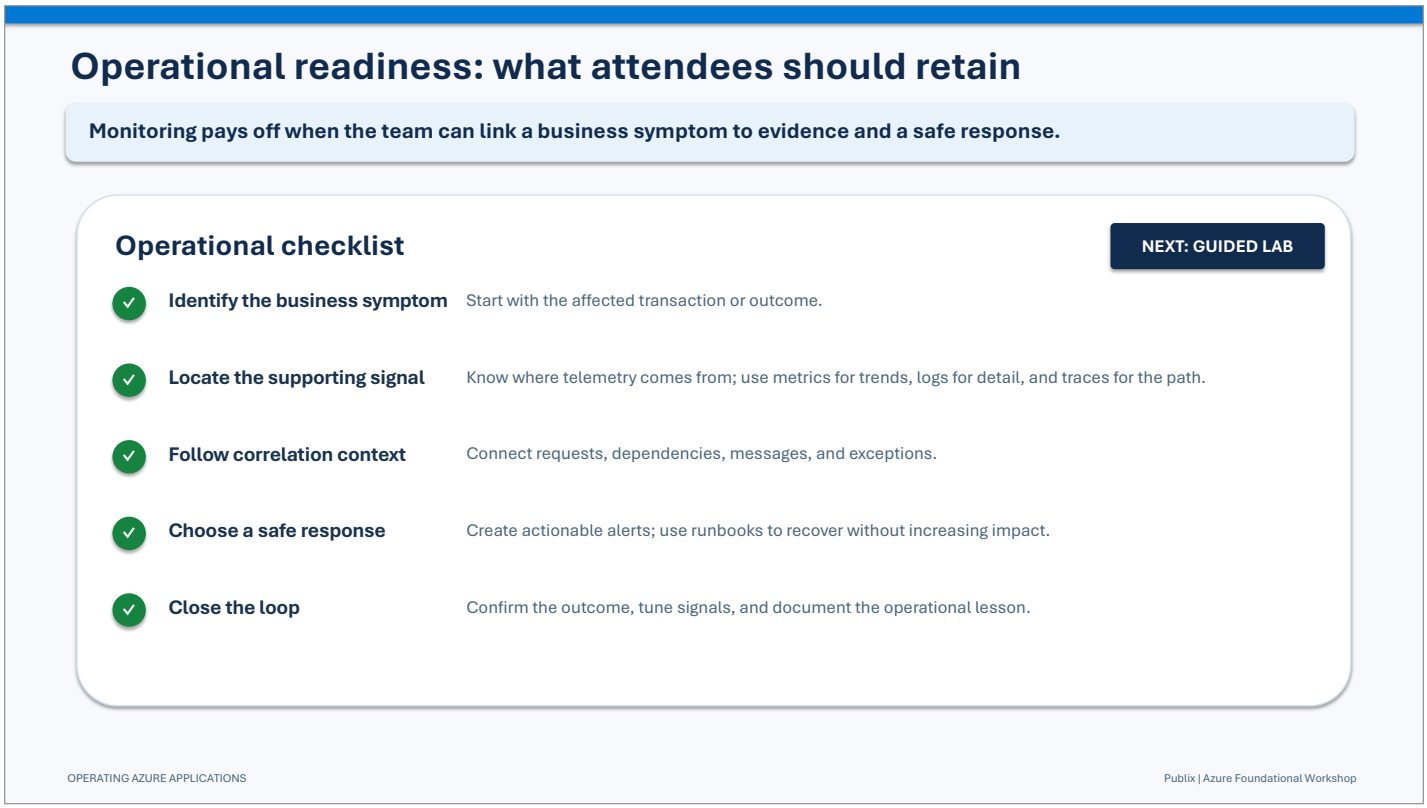
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/action-groups



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These are recommended categories rather than a guaranteed out-of-the-box dashboard. Business outcome metrics may require custom instrumentation. Service Bus and Functions expose different metrics; validate the selected alert signals and data collection settings.

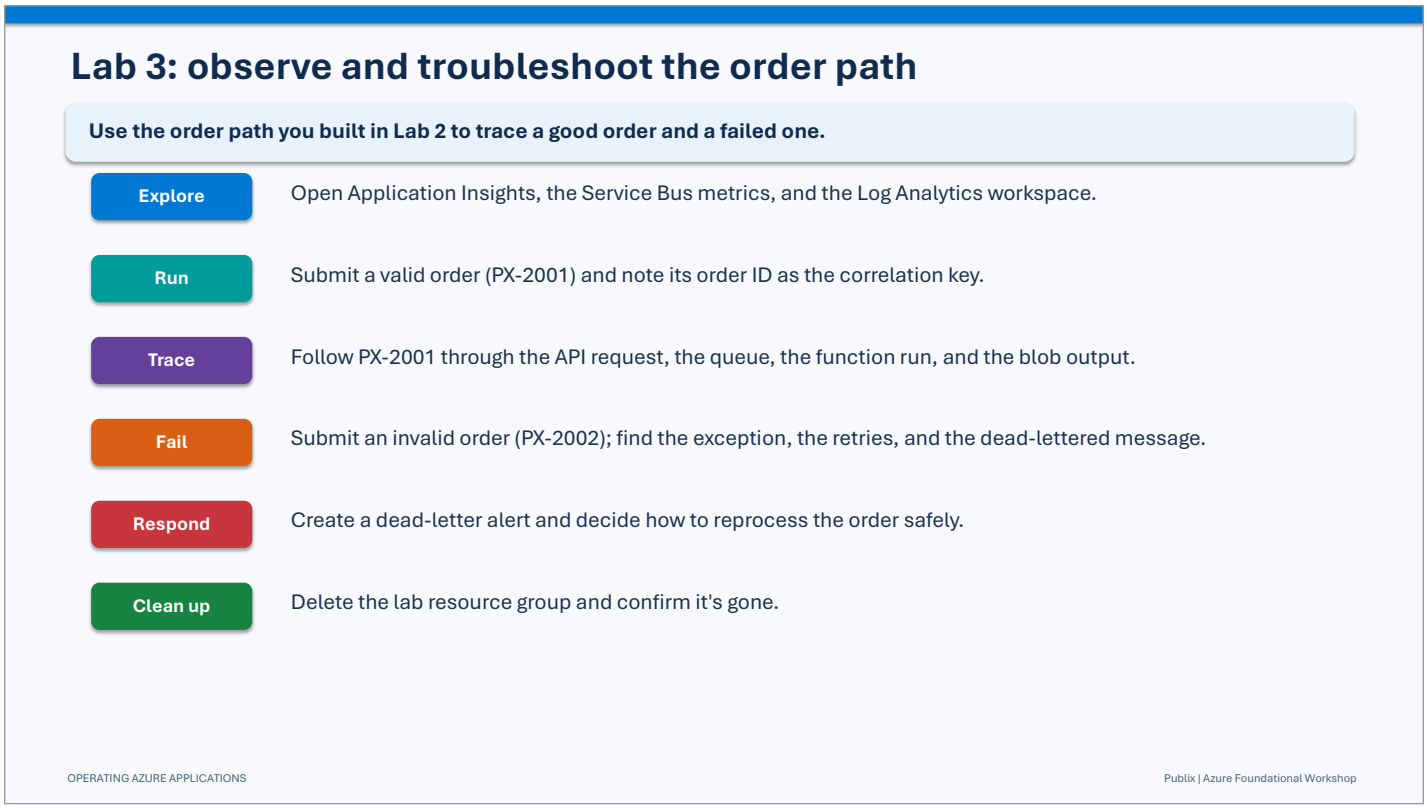
[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview



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End with the mental model: collect useful telemetry, find the affected transaction, diagnose, respond, and improve the system.

[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview



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Lab guide: Operating Azure Applications - Lab 3. Attendees use the order path they built in Lab 2: submit a valid order and follow it end to end, then submit an invalid order and find the exception, the retries, and the dead-lettered message. They create a dead-letter alert, discuss safe reprocessing, and delete the lab resource group. Cost reporting can lag deletion; that's expected.

[Sources]
- https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview
- https://learn.microsoft.com/en-us/azure/service-bus-messaging/explorer
- https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-metric-alert-rule
- https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/delete-resource-group